Attendance Shortage Predictor

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Council for Skills and Competencies

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CERTIFICATE OF COMPLETION

This is to certify that Ms./Mr. ANCHA SWATHI,

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affiliated with JNTU Kakinada, bearing roll number 23761A1269,

has successfully completed the Internship Program in Data Structure and Algorithms using Java,

offered by **Council for Skills and Competencies (CSC India)** Under APSCHE - Student Internship Initiative during the period from 04-05-2026 to 27-06-2026

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CERTIFICATE OF INTERNSHIP

Why I built this project

Keeping track of attendance across different subjects can get confusing. Sometimes we know our percentage, but we still aren't sure how many more classes we need to attend.

I wanted to build a simple tool that shows the current situation clearly and helps students plan ahead.

The problem

Manual calculations take time, especially when several subjects are involved.

The idea

Enter attendance once and get an updated percentage and shortage status.

The goal

Make attendance easier to understand, not just another list of numbers.

What the Website meant to do

Add subjects

Save a subject name, total classes, and classes attended.

Show percentages

Calculate attendance automatically whenever details change.

Flag shortages

Highlight subjects that are below the 75% target.

Predict next steps

Estimate how many consecutive classes are needed to reach the target.

Tools and concepts used

Java

Handles the attendance calculations and application logic.

Java Swing + AWT

Creates the desktop window, buttons, colours, and progress bars.

HashMap

Keeps subject objects indexed by name for quick lookup.

Eclipse IDE

Used to write, run, and debug the Java code.

JProgressBar

Shows attendance visually instead of only as a percentage.

In-memory data

Subjects are kept while the app is open; permanent storage is future work.

How it works

1 → 2 → 3 → 4 → 5

Enter details

Subject name, total classes, classes attended

Calculate

Convert attended/total into a percentage

Check target

Compare the percentage with 75%

Predict

Calculate additional consecutive classes needed

Update

Record the next class and refresh the dashboard

What the student sees

A subject card with attendance percentage, a progress bar, status colour, and actions to predict or record a class.

The calculation behind the prediction

Attendance percentage

(Classes attended ÷ Total classes) × 100

Classes needed

Find the smallest x where (A + x) ÷ (T + x) × 100 ≥ 75%

Example: 30 attended out of 50 classes

Current

30 / 50 = 60%

After 30 classes

60 / 80 = 75%

Prediction

Attend 30 more

Where DSA fits in

HashMap

Each subject name is used as a key. That makes finding a subject quick on average.

Iteration

The app loops through stored subjects to build cards and count eligible/shortage subjects.

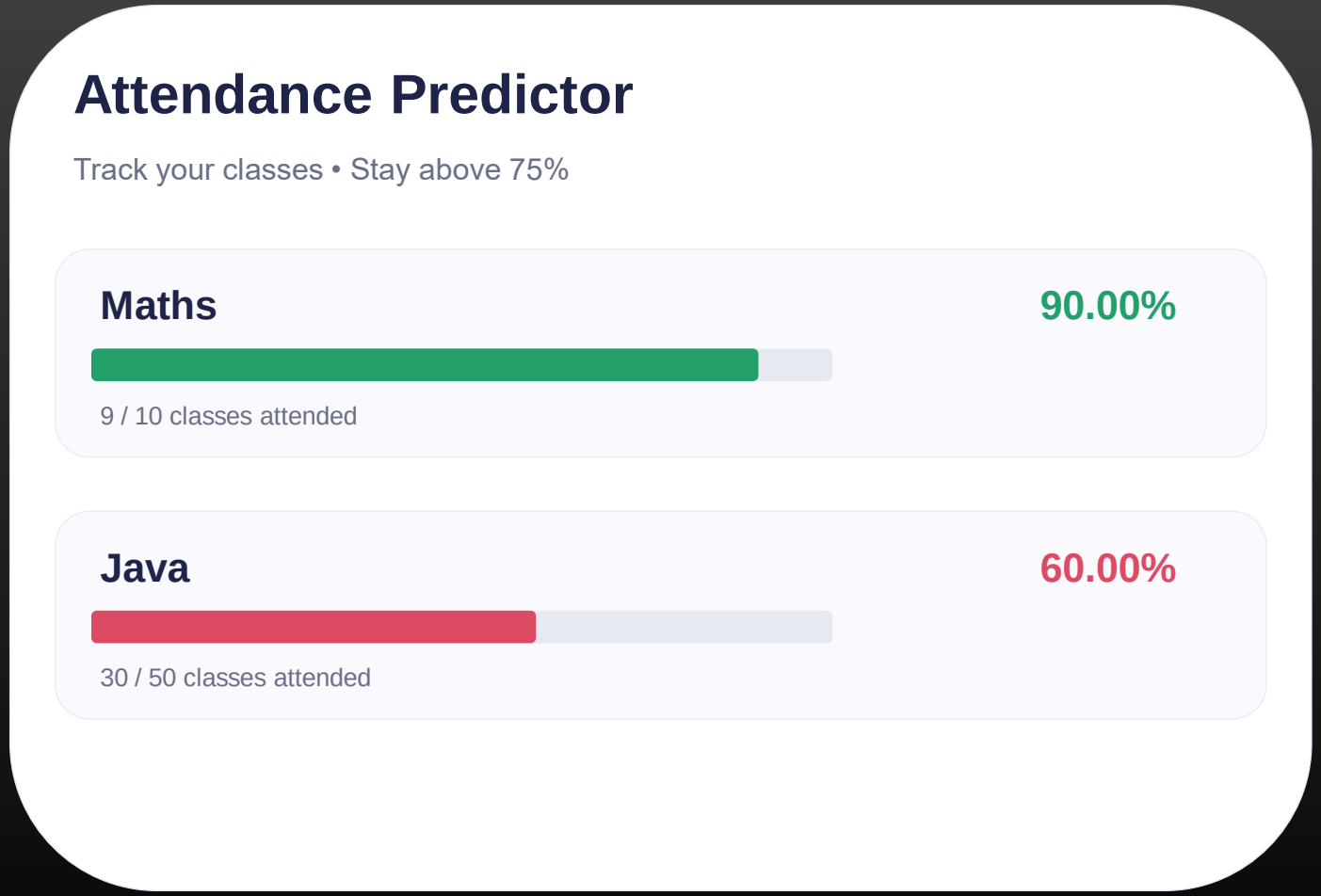
Time complexity

Subject lookup: O(1) average. Building the dashboard: O(n) for n subjects.

Validation

Checks prevent negative class counts, attended classes above total, and duplicate names.

What the interface shows



Progress bars

Quickly compare subjects against the target.

Colour cues

Green for target reached; red for shortage.

Quick actions

Predict classes or record the next class.

Testing and current limitations

Tests to run

- Add a valid subject
- Check percentage and status
- Predict classes needed
- Record present/absent
- Reject invalid class counts

Current limitation

The app stores subjects in memory. Closing the window clears the data, so it is best for a demo until persistence is added.

What works now

Subject entry, percentage display, colour-coded cards, prediction, and attendance updates.

Next improvement

Save data to a file or MySQL so attendance is available after restarting the app.

Thank You

By: Swathi Ancha